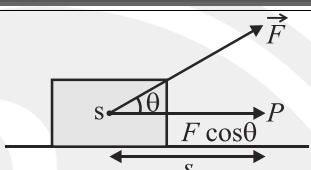
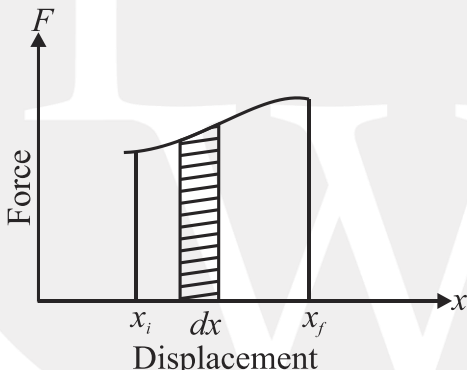
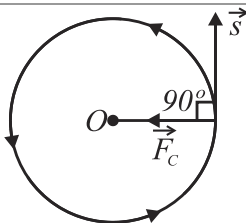
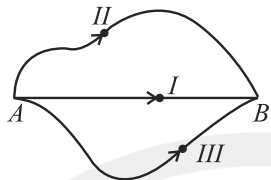


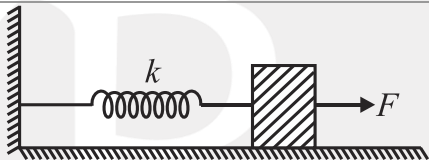
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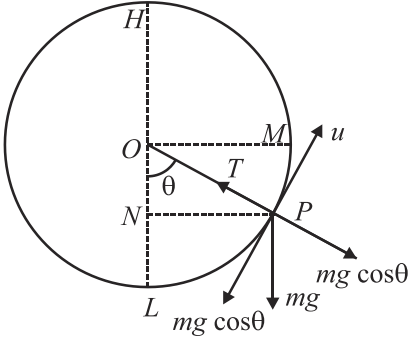
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Work, Energy and Power

Sr. No.	Concept	Formulas	Other information
1.	Work done by constant force (W)	$W = Fs \cos \theta$ $W = \vec{F} \cdot \vec{s}$ 	s = displacement caused by the force F θ = angle between applied force and displacement.
2.	Work done by variable force	$W = \int dW = \int \vec{F} \cdot d\vec{s}$  $W = \int_{x_i}^{x_f} dW = \int_{x_i}^{x_f} F dx$ $W = \int_{x_i}^{x_f} dW = (\text{Area under the curve})$	ds = small displacement. x_i and x_f = the initial and final position respectively. dx = small displacement of the curve.
3.	Work done on a particle moving in three-dimensional space	$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$ $\vec{s} = s_x \hat{i} + s_y \hat{j} + s_z \hat{k}$ $W = \vec{F} \cdot \vec{s}$ $= (F_x \hat{i} + F_y \hat{j} + F_z \hat{k}) \cdot (s_x \hat{i} + s_y \hat{j} + s_z \hat{k})$ $W = F_x s_x + F_y s_y + F_z s_z$	F_x, F_y, F_z = the components of force along x, y, z axis respectively. s_x, s_y, s_z = the components of displacement along x, y, z axis respectively.

Sr. No.	Concept	Formulas	Other information
4.	Work done by centripetal force	The angle between $\vec{F}$ and $\vec{s}$ is 90°. $W = Fs \cos 90^\circ = 0$ F_c is the centripetal force acting towards the center O.	
5.	Work done by conservative forces	Work done by conservative force is path independent.  $W_I = W_{II} = W_{III} = \int F dL$ $\Rightarrow \int_A^B F dL = \int_A^B F dL = \int_A^B F dL$ Path-I Path-II Path-III	W_1, W_2 and W_3 = denote the net work done in moving a body from A to B along three different paths I, II and III respectively.
6.	Kinetic energy of a body	If the body is moving with velocity v : $K = \frac{1}{2}mv^2 = \frac{p^2}{2m} \quad \therefore p = \sqrt{2mK}$	K = the kinetic energy of the body of mass m. p = linear momentum of the body
7.	Work-energy theorem	The work done by the net force acting on a body is equal to the change in kinetic energy. $W = \text{Change in K.E.} = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$	m = mass of the body. u = initial velocity of the body v = final velocity of the body
8.	Potential energy	For a conservative force in one dimension, the potential energy function $U(x)$ is defined as $F(x) = -\frac{dU(x)}{dx}$ Or $\Delta U = U_f - U_i = -\int_{x_i}^{x_f} F(x) dx$ Gravitational potential energy; $U = mgh$	x_f = final position of the body x_i = initial position of the body. U = the gravitational P.E. of a body of mass m at a height h
9.	Mechanical energy conservation	If the net external force acting on a system is zero, then the mechanical energy is conserved. $K_f + U_f = K_i + U_i$ U_f and U_i are the final and initial potential energy respectively	K_f and K_i = the final and initial kinetic energy respectively.

Sr. No.	Concept	Formulas	Other information
10.	Equilibrium conditions	$\vec{F} = -\frac{dU}{dr}\hat{r}, \quad \vec{F} = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$ (i) Stable Equilibrium: $F(r) = -\frac{\partial U}{\partial r} = 0; \text{ therefore } \frac{\partial U}{\partial r} = 0;$ and $\frac{\partial^2 U}{\partial r^2} > 0$ (ii) Unstable Equilibrium : $F(r) = -\frac{\partial U}{\partial r} = 0; \text{ therefore } \frac{\partial U}{\partial r} = 0;$ and $\frac{\partial^2 U}{\partial r^2} < 0$	U = potential energy of system
11.	Potential energy of a spring	 Restoring force $F \propto x$ or $F = -kx$ $W = \int_0^x kx \, dx = \frac{1}{2} kx^2$ This work done is stored as potential energy U of the spring $\therefore U = \frac{1}{2} kx^2$	W = the work done in stretching the spring through distance x . k = the force constant of spring.
12.	Power	Power = $\frac{\text{Work}}{\text{Time}}$ or $P = \frac{W}{t}$ Instantaneous power is given by $P = \frac{dW}{dt} = \frac{d}{dt}(\vec{F} \cdot \vec{s}) = \vec{F} \cdot \frac{d\vec{s}}{dt}$ $= \vec{F} \cdot \vec{v} = Fv \cos \theta$ If $\theta = 0^\circ$, then $P = Fv$.	P = power delivered by the body. F = net force applied on the object. v = instantaneous velocity of the object.

Sr. No.	Concept	Formulas	Other information
13.	Motion in a vertical circle	 Minimum velocity at the lowest point for looping the vertical loop, $v_L = \sqrt{5gr}$ Velocity at the highest point for looping the loop, $v_H = \sqrt{gr}$ Tension at P : $T = \frac{mu^2}{r} + mg \cos \theta$ Tension at the lowest point (Maximum tension) : $T_L = \frac{mv_L^2}{r} + mg$ Tension at the highest point (Minimum Tension) : $T_H = \frac{mv_H^2}{r} - mg$	r = the radius of the vertical circle. u = the linear velocity of body at P. g = the gravitational acceleration.